

Class 8 Physics Sound

Introduction

Have you heard the following sounds - Sitar, Violin, Veena, your favorite song playing in your iPod, temple bells, chirping of birds, vehicles honking in traffic, dogs bark, pressure cooker whistling, crackers bursting, rattling and breaking of objects etc. While some sounds may be pleasant and soothing, some could be annoying. But all these are **Sounds**. In this chapter, we will study about these Sounds, how we speak, how we hear, their characteristics and so on



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Sound is produced by a vibrating body

Sound is produced by a **vibrating body**. In other words, any vibrating body would always produce sound. Sometimes, the vibration is clearly visible and the sound is strong, whereas in some cases, the vibration is small and the sound is feeble. (Note : The to and fro or back and forth motion of a body is termed as **vibration or oscillatory motion**). When the diaphragm of a drum is struck, it vibrates and produces sound. When the strings in a violin vibrates, it produces sound. When the air column vibrates within a flute, it produces sound. As seen, **different parts** of the musical instruments (diaphragm of the drum, strings of violin, air column within in flute) vibrate to produce sound.



Drums



Violin



Flute

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The Voice Box

In humans also, sound is produced because of the vibration of his **voice box** also known as **larynx**. The Larynx is situated at the upper end of **windpipe**. It comprises of **2 stretched membranes** called **vocal cords** with a narrow **slit** between them. When we talk, **air passes through** those stretched membranes they produce different of **sounds** in accordance with the stretching



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Sound needs a medium for propagation

Sound can propagate through **solids, liquids and gases**. In fact, it needs a medium to propagate, be it solid, liquid or gases. **It cannot propagate through vacuum.**

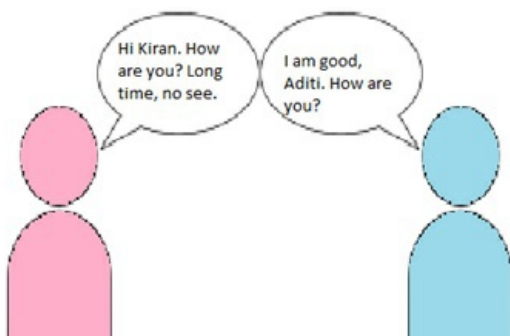
Sound can propagate through Solids: The doctor checks us using the stethoscope. He can hear the sound from within our body because sound travels through solids



Sound can propagate through liquids: In fact, this is how aquatic animals communicate with each other.



Sound can propagate through gases/air: This is the reason why we are able to hear a person standing near us speak.



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The Human Ear



The shape of our outer ear is like **a funnel**. Sound waves enter inside. It travels down a canal and at the end of it, touches a thin stretched membrane called the **eardrum**. The eardrum vibrates to the effect of the sound. The ear passes on these vibrations to the **inner ear** and then there to the **brain**, which processes it. That is how we hear.!

To understand better, take a toy drum. Bang on diaphragm with various intensity blows. First strike gently with your hands. Then blow slightly harder with the cords. Then very hard on so. In all the cases, see the diaphragm vibrates according to the intensity of the blow and then it produces the sound. This is analogous to the ear.

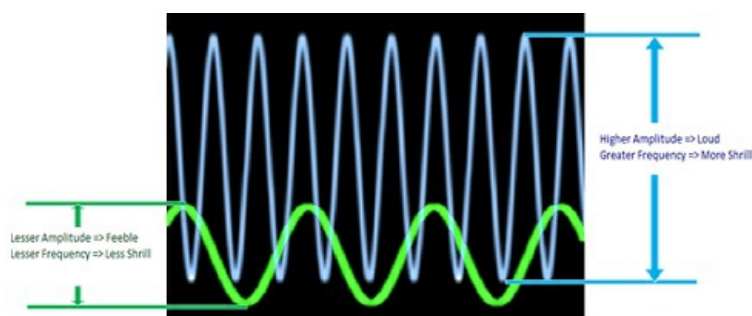


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Characteristics of Sound

Sound is characterized by 2 important features

- **Frequency**
- **Amplitude**



Frequency : The number of **oscillations per second** is called the frequency of oscillation. Frequency is expressed in **hertz**. Its symbol is **Hz**.

Frequency = $\frac{\text{No of oscillations}}{\text{Time (in seconds)}}$

If pendulum oscillates 15 times in one minute, the frequency is given by

$$\text{Frequency} = \frac{\text{No of oscillations}}{\text{Time (in seconds)}} = \frac{15}{30} = 0.5 \text{ Hz}$$

Frequency determines the **shrillness or the pitch of the sound** **Higher the frequency, higher the pitch or it is said that the sound is shrilling. Lower the frequency, lower the shrill or pitch.** Eg: Babies voice has more frequency and thereby more shrilling than an adult voice.

Amplitude: Amplitude is a measure of the **intensity of the sound** Amplitude determines the **Loudness** of sound. Loudness is proportional to the **square of the amplitude** of the vibration body producing sound

Loudness = $k \cdot (\text{Amplitude})^2$ (where k is the proportionality constant)

The loudness is expressed in a unit called **decibel (dB)**.

The more the amplitude of vibration, the sound is the loud. The lesser the amplitude of vibration, the sound is feeble.

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Audible and Inaudible sounds

- The human ear can detect sounds in the range of **20Hz-20KHz**. These are called Audible Sounds. Eg: Music, Instruments, Crackers.
- Frequencies **below 20Hz and greater than 20KHz cannot be heard by the human ear** These are called Inaudible Sounds. Eg: It is believed that bat calls are in the inaudible ultrasonics range. Dogs can hear frequencies greater than 20KHz, so they are used by the police to respond to such whistles. Ultrasound scans produce sounds in this range.



Singer Singing a song



Bat calls in inaudible range

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Noise and Music

- **Music** refers to **pleasant sounds** which we would like to hear. Eg: Musical instruments like flute, guitar, violin etc. Your favourite singer singing a melodious song.
- Sounds which are **unpleasant to hear** are termed as **noise**. Eg : Honking of vehicles, Loucrackers, Explosions



Music from instrument



Noise due to explosions

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Noise Pollution

Presence of excessive or unwanted sounds in the environment around us is called as **Noise Pollution**.

Sources of Noise Pollution

- Vehicles honking on the road and traffic signals
- Explosions
- Bursting crackers
- Huge machines at construction sites, factories
- Loud Speakers
- TV, Music system played at high volumes
- Some kitchen and home appliances like mixers, ACs etc



Loudspeakers



Crackers

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Harms of Noise Pollution

Noise pollution mostly leads to **health related problems** such as

- Anxiety, Stress
- Hypertension (High BP)
- Lack of sleep
- Prolonged exposure can lead to temporary or even permanent hearing impairment



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Measure to limit Noise Pollution

Measure to limit Noise Pollution

- Industry and factory units which produce a lot of noise should be setup away from residential areas.
- Silencers (Silencing devices) should be installed in air crafts, industry and home appliances which tend to produce a lot of noise.
- A lot of trees should be built around such factory units so that they absorb sound and protect the residential areas around them from the loud noise.
- Minimal use of horns in vehicles. No honking.
- Low volume for TV, music systems at home.
- Avoid or minimize use of crackers.



No horns zone



Low Volume for TVs